

# Introduction to Radiobiological Models



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- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)

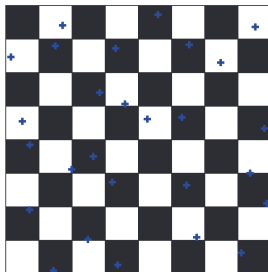
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- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

# Mathematical Modeling

## Poisson Statistics



**Figure:** If it rains on a chessboard, and an average of  $x$  drops hits each square, How many squares are dry?

- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

# Mathematical Modeling

## Poisson Statistics

- If  $x$  is the average number of drops per square, then the probability of a square being hit by  $k$  drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

# Poisson Distribution

## Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of  $x$  lethal hits per cell:
    - $e^{-x}$  cells survive (no hits).
    - $1 - e^{-x}$  cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells.  $SF = 0.37$
    - 2 hits kills 86% of cells.  $SF = 0.14$
    - 3 hits kills 95% of cells.  $SF = 0.05$
    - 4 hits kills 98% of cells.  $SF = 0.02$
    - 5 hits kills 99% of cells.  $SF = 0.01$

### Note

$D_0$  is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

# Poisson Distribution

## Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of  $x$  surviving tumor cells per patient:
    - $e^{-x}$  cells survive (no hits).
    - $1 - e^{-x}$  cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.:  $TCP = 0.37$
    - 2 tumor cell per pt.:  $TCP = 0.14$
    - 3 tumor cells per pt.:  $TCP = 0.05$
    - 4 tumor cells per pt.:  $TCP = 0.02$
    - 5 tumor cells per pt.:  $TCP = 0.01$

### Note

To achieve a certain TCP, you should aim for a tumor cell survival of  $(1 - TCP)$ .



# Target theory

## Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

where  $D$  is the dose and  $D_0$  is the dose required to reduce the surviving fraction to 37%.

- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

# Target theory

## Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose  $D$ , the survival fraction (SF) is given by:

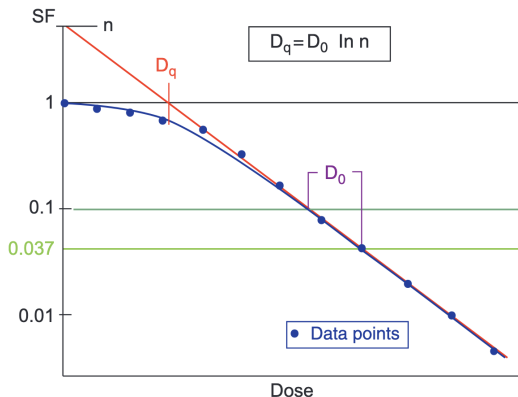
$$SF = 1 - (1 - e^{-D/D_0})^n$$

where  $D$  is the dose,  $D_0$  is the dose required to reduce the surviving fraction to 37%, and  $n$  is the number of targets.

- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

# Target theory

## Multitarget, Single-Hit Model



**Figure:** The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate  $D_0$  by taking measurements in the high-dose region

# Target theory

## Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - $D_0$  is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

### Important to note

Do not measure  $D_0$  at  $SF = 0.37$  if the survival curve has a shoulder. Remember that  $D_0$  applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between  $SF = 0.1$  and  $SF = 0.037$ .

- $D_0$  is a measure of the cell's inherent radiation sensitivity. Most  $D_0$  values are somewhere around 1 Gy.
- $D_{10} = 2.3 \times D_0$  = a dose that will reduce surviving fraction by tenfold (i.e., from  $SF = 0.1$  to  $SF = 0.01$ ).
- The low-dose portion of the curve is known as the “shoulder”:
  - $D_q$  tells you how wide the shoulder is.
  - $D_q$  is a measure of the cell's repair capacity. More repair means a larger  $D_q$  and a larger shoulder

# Single-Hit, Multitarget Model

## Advantages and Disadvantages

- Advantages:

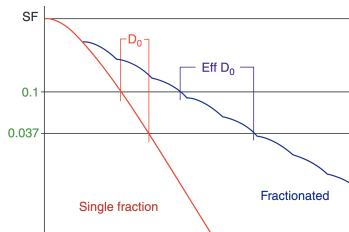
- The high-dose portion of the curve is a straight line.
- This straight-line component correlates well with cell-culture experiments.

- Disadvantages:

- The low-dose portion of the curve greatly underestimates cell kill.
- Unlike the linear-quadratic model, the single-hit model is not based on a molecular mechanism.
- Unlike the linear-quadratic model, there is not a simple equation for “equivalent dose” given a daily fractionation schedule.

# Fractionated Radiation and Effective $D_0$

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- $D_0$  versus effective  $D_0$ . When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective  $D_0$  is always larger than  $D_0$



# Fractionated Radiation and Effective $D_0$

Continue

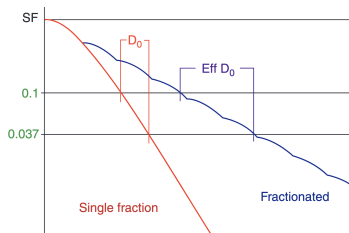
- The slope of a fractionated survival curve is called effective  $D_0$ .
- For a single fraction of  $D$  Gy with a surviving fraction of  $SF_D$ :

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$

- After  $x$  fractions to total dose  $xD$  Gy:  
 $SF_x^D = e^{-xD/\text{effective } D_0}$



# Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose  $D$ .
- Lethal DNA aberrations  $= \alpha D + \beta D^2 = \text{Cell Kill}$

$$SF_D = e^{-(\alpha D + \beta D^2)}$$

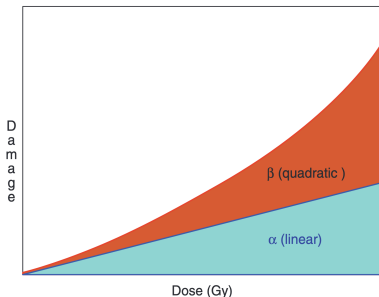
- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:
  - Single-hit kill ( $\alpha$ ) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.
  - Multi-hit kill ( $\beta$ ) is repairable damage and is **dependent** on fractionation or dose rate. This corresponds to double-hit and **inter**-track accumulated damage.



# Linear-Quadratic (LQ, Alpha-Beta) Model

Continue:

- The  $\alpha/\beta$  ratio is the **dose** at which  $\alpha$  kill and  $\beta$  kill are equal.
  - Low  $\alpha/\beta$  ratio (“high repair”) tissues are relatively resistant at small fraction size and relatively sensitive at large fraction size.
  - High  $\alpha/\beta$  ratio (“low repair”) tissues are relatively sensitive at small fraction size and relatively resistant at large fraction size.



# Linear-Quadratic (LQ, Alpha-Beta) Model

## Alpha-Beta Model and Dose Fractionation

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have  $n$  no of fractions of dose  $d$  each, then the total dose  $D = nd$ . The surviving fraction after  $n$  fractions is given by:

$$SF_n = e^{-n(\alpha d + \beta d^2)}$$
$$\Rightarrow -\frac{1}{nd} = \frac{\alpha}{\ln(SF_n)} + \frac{\beta d}{\ln(SF_n)}$$

- Plot of dose per fraction  $d$  versus  $-\frac{1}{nd}$  gives a straight line with slope  $\beta/\ln(SF_n)$  and y-intercept  $\ln(SF_n)$ . The ratio of intercept to slope gives  $\alpha/\beta$ .
- Such values are obtained for a variety of tissues and tumors, and are used to guide clinical decision-making regarding fractionation schemes.

# Linear-Quadratic (LQ, Alpha-Beta) Model

## $\alpha/\beta$ Ratios of Tissues and Tumor

- Most acute-reacting tissues are believed to have an  $\alpha/\beta$  ratio  $\approx 10$ .
- Most late-reacting tissues are believed to have an  $\alpha/\beta$  ratio  $\approx 3$  or less.
- CNS tissue (brain, cord) is believed to have an  $\alpha/\beta$  ratio  $\approx 1-2$ .
- Most tumors are believed to have an  $\alpha/\beta$  ratio  $\geq 10$ .
- Some low  $\alpha/\beta$  tumors may have an  $\alpha/\beta$  ratio  $\approx 1.5-4$ . Breast and Prostate are the “classic” low  $\alpha/\beta$  tumors.

# Linear-Quadratic (LQ, Alpha-Beta) Model

## Biologically Effective Dose

- One advantage of the linear-quadratic model is that it is easy to calculate effective doses for different fraction sizes.

### Definition

Biologically effective dose (BED) is an extrapolated dose given over infinitely many fractions.

- For  $n$  fractions of  $d$  dose per fraction:

$$\begin{aligned}\text{BED}_{\alpha/\beta} &= nd \left( 1 + \frac{d}{\alpha/\beta} \right) \\ &= D \left( 1 + \frac{d}{\alpha/\beta} \right)\end{aligned}$$

- $\text{BED} = \text{total physical dose} \times \text{relative effectiveness of that dose.}$

# Treatment Interruption

## Consequences

Explain the consequences of unplanned gaps on tumour cells during radiotherapy treatment.

- **Repopulation** of tumour cells may occur.
- The unplanned gap during radiotherapy will cause the cells within a **tumour to proliferate** and leads to an increase in the number of cells that the radiation needs to kill.
- There is evidence that **repopulation can be accelerated in the treated tumours** so that this effect can be significant.
- Thus an increase of overall treatment time (due to the gap) will decrease the efficacy of radiotherapy.

# Treatment Interruption

## Recommendations

Give the recommendations about how to compensate for unscheduled gaps in radiotherapy treatment.

- 1 The main recommendation is to compensate for a gap by **continuing to deliver the same number of fractions in the same overall time**, by inserting the missed fractions (using the original dose per fraction) either during weekends or as extra fraction on previously treatment days.
- 2 When multiple fractions are given in a day, the interval between them should be **at least 6 hours**.
- 3 If the break in treatment occurs early in the schedule, then two fractions per day can be used. However, if there are not enough days left on the original schedule to catch up using two fractions on each remaining days, three fractions per day can be considered subjected to the availability of resources.

**Question 1:** Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

- ① Total Dose
- ② Dose per fraction
- ③ **BED**
- ④ Relative effectiveness of that dose

# MCQ Example

**Question 2:** A tumour containing  $10^8$  cells is inactivated by radiation with a mean lethal dose of  $D_0 = 1.8$  Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

- ①  $1.2 \times 10^{-3}$
- ②  $1.4 \times 10^{-4}$
- ③  $1.4 \times 10^{-3}$
- ④  $1.2 \times 10^{-4}$



## MCQ Example

**Question 2:** A tumour containing  $10^8$  cells is inactivated by radiation with a mean lethal dose of  $D_0 = 1.8$  Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

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- ③  $1.4 \times 10^{-3}$
- ④  $1.2 \times 10^{-4}$

### Solution

The number of surviving cells is given by:

$$N = N_0 e^{-D/D_0}$$

where  $N_0$  is the initial number of cells,  $D$  is the dose, and  $D_0$  is the mean lethal dose. Substituting the values, we get:

$$N = 10^8 e^{-45/1.8} \approx 1.4 \times 10^{-3}$$

# Linear-Quadratic (LQ, Alpha-Beta) Model

## Practical Applications

### Problem 1

If a treatment is planned for 70 Gy/ 35 fractions, but due to departmental constraints, we need to complete it in 25 fractions. What is the new dose per fraction for same tumor control with  $\alpha/\beta = 10$ ? (Ans: 2.65 Gy)

# Linear-Quadratic (LQ, Alpha-Beta) Model

## Equivalent Dose in 2 Gy fractions (EQD<sub>2</sub>)

- EQD<sub>2</sub> is the dose in 2 Gy fractions that would produce the same biological effect as the actual treatment.
- For n fractions of d dose per fraction:

$$\text{EQD}_2 = \frac{BED}{1 + \frac{2}{\alpha/\beta}}$$
$$\Rightarrow \text{EQD}_2 = D \left( \frac{d + \alpha/\beta}{2 + \alpha/\beta} \right)$$

# Linear-Quadratic (LQ, Alpha-Beta) Model

## Practical Applications

### Problem 2

You want to treat a lung cancer to 60 Gy in 2 Gy fractions, but the first five fractions are given at 3 Gy/fx due to SVC syndrome. Assuming  $\alpha/\beta = 3$ , how much additional dose should you deliver at 2 Gy per fraction?  
(Ans: 42 Gy)

# Linear-Quadratic (LQ, Alpha-Beta) Model

## Practical Applications

### Problem 3

A 40-year-old lady with colorectal cancer is being treated with radiotherapy. A dose of 45 Gy in 25 daily fractions over 5 weeks is planned. After ten fractions, treatment was cancelled for five fractions due to machine breakdown. It was planned to deliver the isoeffective tumour dose by increasing the size of the ten fractions to finish the treatment as scheduled. By using the "equivalent dose" method, calculate required dose per fraction for the last ten fractions. Assume  $\alpha/\beta = 10$  Gy for colorectal cancer. (Ans: 2.54 Gy)

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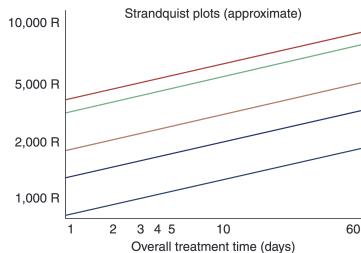
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# Need for time-dose models

- The aim of the radiotherapist is to obtain high probability of tumor control with minimal normal tissue complications.
- Since typical treatment plans are quite complicated to compare so with the help of this models. the treatment regimes can be easily modified.
- Time dose relationship have been evolved in different forms like graphical, tabular and mathematical forms. The most commonly used models are:
  - Nominal Standard Dose (NSD)
  - Cumulative Radiation Effect (CRE)
  - Time Dose Fractionation (TDF)

# Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:
  - Back in the 1930s-1940s, Strandquist treated a bunch of skin cancers and plotted radium exposure versus skin erythema, desquamation, necrosis, and tumor cure.
  - He found that the relationship between total dose and overall treatment time formed a series of parallel lines on a logarithmic chart.
  - The Strandquist plots (approximate). These parallel lines represent various skin-related endpoints such as erythema, desquamation, and necrosis.





# Nominal Standard Dose (NSD)

## Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric “**curve fitting**” model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.
- The NSD makes no attempt to predict late toxicity.
- There are two versions of the equation: one with time and fractionation, another with just fractionation.
- For  $N$  fractions of total dose  $D$  (in rads) delivered over  $T$  days:

$$D = \text{NSD} \times N^{0.24} \times T^{0.11}$$

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# Cumulative Radiation Effect (CRE)

- NSD formula does not say anything about tumor response and normal tissue tolerance due to part of the treatment schedule which had already been completed, might have reached some level of tolerance.
- Kirk et. al. extended the NSD concept to the cumulative radiation effect (CRE) for such type of schedules. This is based on sub-tolerance of normal tissue which was subjected to irradiation.
- The partial tolerances are additive in nature,

$$PT = \sum PT_i$$

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# Time-Dose-Fractionation (TDF)

- Orton and Ellis(1973) developed time dose and fractionation concept to make use of partial tolerance more practicable. The TDF for fractionated beam therapy was given by:

$$\text{TDF} = d^{1.538} \times n \times x^{-0.169} \times 10^{-3}$$

where  $d$  = dose per fraction,  $n$  = number of fractions,  $x = t/n$ ,  $t$  = overall treatment time in days.